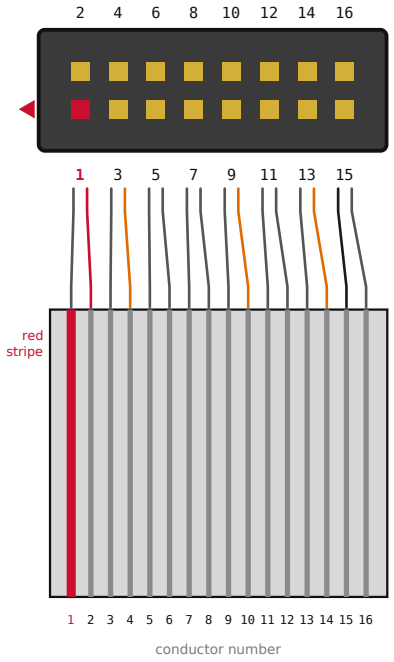
J1 BOX HEADER ↔ 16-WAY RIBBON ↔ DIP-16 IDC PLUG — continuity check sheet

Standard DIP-IDC plug: conductors 1,2 land on DIP pins 1,16; 3,4 on 2,15; … 15,16 on 8,9. TO BE CONFIRMED on YOUR cable with a meter before first power-up (mandatory).

J1 box header (top view, as on the PCB)

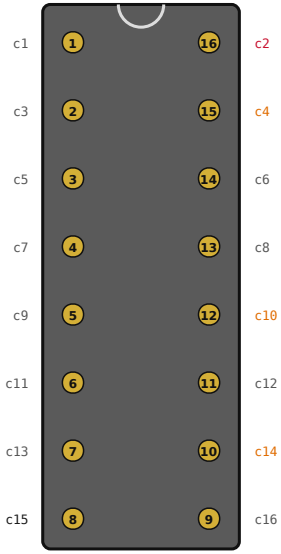


16-way ribbon, 1.27 mm (drawn ×2 wide)

IDC socket at J1: conductor n → J1 pin n (red stripe → pin 1).
J1: top row even pins 2…16, bottom row odd pins 1…15; ◄ = pin 1.

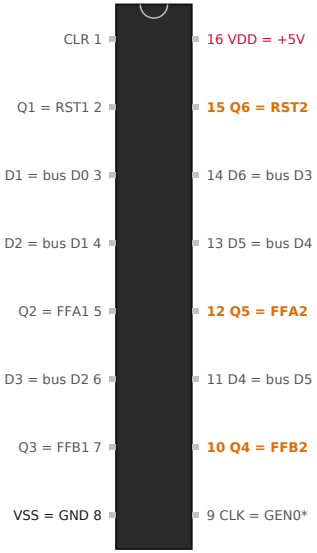
DIP-16 IDC plug: conductor → pin

(plug seen from the cable side, notch/pin 1 at the top)



Each row of the plug takes two neighbouring conductors: odd n → left pin, even n → right pin.

In the DX U701 socket (top view, notch up)



MAPPING TABLE (source: hardware/kicad/tools/netlist.py)

c / J1	DIP	U701 signal	expect on the board
1	1	CLR (tied +5V)	U1 pin 1
2	16	VDD = +5V	U1 pin 16, +5 V
3	2	Q1 = RST1 (bass)	U1 pin 2
4	15	Q6 = RST2 (SNARE)	→ U3 pin 4 (mux Y)
5	3	D1 = bus D0	U1 pin 3
6	14	D6 = bus D3	U1 pin 14
7	4	D2 = bus D1	U1 pin 4
8	13	D5 = bus D4	U1 pin 13
9	5	Q2 = FFA1 (bass)	U1 pin 5
10	12	Q5 = FFA2 (SNARE)	→ U3 pin 7 (mux Y)
11	6	D3 = bus D2	U1 pin 6
12	11	D4 = bus D5	U1 pin 11
13	7	Q3 = FFB1 (bass)	U1 pin 7
14	10	Q4 = FFB2 (SNARE)	→ U3 pin 9 (mux Y)
15	8	VSS = GND	U1 pin 8, TP4 GND
16	9	CLK = GEN0*	U1 pin 9

LEGEND

- intercepted snare lines (DIP 10, 12, 15)
- +5 V (DIP 16); red stripe = conductor 1
- GND (DIP 8)
- straight-through signals

CONTINUITY TESTS (meter on beep / Ω; A–D: all chips and the Nano OUT of their sockets)

A Cable alone. Probe J1-socket hole n (or the cable's pin-1 end) and the DIP plug: conductor n must beep ONLY to the DIP pin in the table (c1→1, c2→16, c3→2 … c16→9). Also check each DIP pin against its two neighbours: no beep (catches a squashed IDC).

B Cable plugged into J1 on the TimeShift. DIP plug pin k ↔ U1 socket pin k must beep for k = 1–9, 11, 13, 14, 16 (13 straight-through pins).

C DIP plug pins 15 / 12 / 10 ↔ U3 socket pins 4 / 7 / 9 must beep, and must NOT beep to U1 pins 15 / 12 / 10 (those go to the Nano and the mux A inputs only).

D DIP pin 16 (+5 V) ↔ DIP pin 8 (GND): no dead short (the meter may beep briefly while C1 charges).

E Plug orientation in the DX socket (docs/03-install-in-the-dx.md §8.1). DX switched OFF and unplugged from the mains, DIP plug fitted in the U701 socket; chips may stay fitted. Meter on Ω: TP4 ↔ DX U702 pin 8: < 1 Ω. C1 + lead ↔ U702 pin 16: < 1 Ω. TP4 ↔ U702 pin 16: NOT near 0 Ω (may climb as capacitors charge). TimeShift U1 pin 1 ↔ C1 + lead: < 1 Ω.
Record A–E in docs/04-test-and-calibrate.md §1. Any mismatch: STOP, do not power the DX.

PROJECT — SHEET 1 / 1 UNITS mm

DX TimeShift (snare) for Oberheim DX, board 1721A

DRAWING — A4 landscape

J1 / ribbon / DIP-16 plug mapping

DWG NO.	SCALE	REV	DATE
DXTS-MECH-003	NTS	A	2026-11-06